Position Sizing as a Risk-Control Variable: A Closed-Loop GRPO Framework for LLM Trading

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***Abstract*—Recent work on reinforcement-learning (RL) fine-tuning of Large Language Models (LLMs) for trading has shown promise, but existing systems entangle reward-driven optimisation with supervised warm-up on expert-curated trading analyses and reduce a trading decision to a single trading action with no position size or exit conditions. We present a closed-loop framework that fine-tunes a generic instruction-tuned LLM into a complete trading agent using only publicly available technical indicators, news, and financial statements, with no supervised cold-start. The action space is a structured strategy specifying action, per-trade quantity, take-profit, and stop-loss; a multi-component reward jointly grades output structure, factual grounding, regulatory compliance, and path-dependent profit; and a two-stage Group Relative Policy Optimization (GRPO) curriculum stabilises the output distribution before optimising profit performance. On a five-ticker, five-month out-of-sample test, pure RL alone lifts both Llama-3.2-3B and Qwen3-4B over their bases and over a value-based DQN baseline. An action-space ablation isolates per-trade quantity as a risk-control variable: removing it recovers Buy & Hold parity on cumulative return at the cost of an approximately three-percentage-point deeper drawdown, characterising explicit sizing not as a performance concession but as a path-stability lever.**

***Index Terms*—large language models, reinforcement learning, GRPO, algorithmic trading, action-space design, risk control**

I. Introduction

Profitable trading requires the rapid synthesis of heterogeneous information — numerical price data and technical indicators alongside unstructured text such as financial news and corporate filings — at a volume that long ago outpaced what manual analysis can sustain. Reinforcement learning (RL) has been a natural fit for this sequential, reward-driven setting, and deep RL architectures such as DQN [1], PPO, and A2C have been widely applied to portfolio management and trade execution, with FinRL [2] standardising the paradigm. Yet recent empirical work continues to find a practical ceiling: a carefully tuned DQN trader [3] cannot consistently outperform conventional algorithmic baselines, pointing to the representational and reasoning limits of traditional RL agents that emit actions as opaque policy decisions with no awareness of the qualitative information human traders routinely use.

Large Language Models (LLMs) open a complementary direction grounded in language understanding and reasoning. The closest recent precedent for this paper, Trading-R1 [4], applies Group Relative Policy Optimization (GRPO) [5] on top of a supervised cold-start. Their published ablation already shows that pure RL meaningfully improves over a non-RL base, although it remains below the SFT-warmed and combined variants in their setup. Building on this observation, we ask a complementary question: can an alternative structural design — a closed-loop action space that includes per-trade quantity, take-profit, and stop-loss, paired with a multi-component reward stack — further amplify the contribution of RL on top of a generic instruction-tuned LLM, without relying on an SFT corpus? Trading-R1's onfold action abstract away position size and exit conditions [6], leaving risk management outside the policy; a richer schema brings it inside the reward signal.

We present a closed-loop framework that addresses both limitations. A generic instruction-tuned LLM is fine-tuned through pure RL, with no SFT cold-start, into a trading agent that emits a structured strategy specifying action, per-trade quantity, take-profit, and stop-loss — making the exit plan and the sizing first-class outputs of the policy. A two-stage GRPO curriculum first stabilises the output distribution under four shaping rewards (format, length, factual grounding, regulation) and then optimises a correctness reward function, the profit oriented optiomizer, over a five-day look-ahead horizon. The framework relies exclusively on publicly accessible data: technical indicators, financial news, and quarterly statements.

We empirically support two claims. **RQ1:** pure-RL fine-tuning lifts both Llama-3.2-3B and Qwen3-4B over their base counterparts and dominates a value-based DQN baseline by a wide margin on every metric reported. **RQ2:** per-trade quantity is a risk-control variable rather than a performance variable — removing it from the action space and retraining recovers Buy & Hold parity on cumulative return at the cost of a deeper maximum drawdown, characterising explicit sizing as a path-stability lever folded into the policy itself.

II. Related Work

Deep RL for trading. Deep RL has been widely applied to trading: DQN [1], PPO [7], A3C [8], and DDPG [9] cover value-based, on-policy, and continuous-control settings, with ensemble methods [10] selecting among them by Sharpe ratio and FinRL / FinRL-Meta [2], [11] providing standard environments. Recent empirical work questions the ceiling of this paradigm: a tuned DQN trader [3] cannot consistently outperform algorithemic trading baselines, attributed to limited representational capacity and the inability of traditional RL agents to ingest qualitative information such as news, fundamentals, and regulatory context. The action space is also typically a discrete buy/sell/hold variable, with size and exit conditions left outside the policy.

LLM agents in finance. BloombergGPT [12] demonstrated that a 50B-parameter model trained on a proprietary financial corpus outperforms general-purpose models on financial NLP tasks but stopped short of direct trading. Open alternatives such as FinGPT [13] and the PIXIU benchmark [14] have since broadened access to financial-domain LLMs and instruction-tuning data. Lopez-Lira and Tang [15] provide an early look at zero-shot LLM-based stock-return forecasting. The most direct precedent for the present work, Trading-R1 [4], combines supervised fine-tuning, reasoning-oriented fine-tuning, and GRPO into a three-stage pipeline. Their action space is a single trading action with professional analysis, and their ablation shows pure RL improving over a non-RL base while still trailing the SFT-warmed and combined variants in that setup — which leaves open whether a richer action space and reward stack can move the pure-RL operating point further.

RL fine-tuning of LLMs. Reinforcement learning from human feedback [16] established the dominant recipe for aligning instruction-tuned LLMs. GRPO [5] replaced the learned value network with an on-the-fly group baseline and was scaled in DeepSeek-R1 [17]; reference-free objectives such as DPO [18] and SimPO [19] target preference data. These methods focus on text-quality alignment; we adapt GRPO to a multi-component reward tailored to closed-loop economic decisions.

Positioning. Our framework explores a complementary structural design to Trading-R1 along three orthogonal axes: (a) the action space is closed-loop with `quantity`, `take-profit`, and `stop-loss` as first-class fields rather than a single trading action; (b) the training signal is a five-component reward shaping output structure, factual grounding, rule compliance, and path-dependent profit jointly rather than a single classification objective; and (c) training is pure RL on publicly accessible data, with no SFT or RFT corpus. Sec. IV-B examines how this alternative design shapes the empirical contribution of pure-RL fine-tuning.

III. Method

A. Framework Overview

The proposed framework is an end-to-end pipeline that turns publicly accessible market data into a fine-tuned LLM emitting complete, risk-aware trading strategies. As shown in Fig. 1, four stages compose the pipeline: (i) daily OHLCV with MACD, RSI, and EMA-7/14 indicators, financial news, and quarterly balance-sheet and income-statement filings are aggregated into rolling 7-day observation windows; (ii) each window is paired with a synthetically sampled portfolio and account state, and a fixed system prompt declares the JSON output schema and the trading rules; (iii) two-stage GRPO

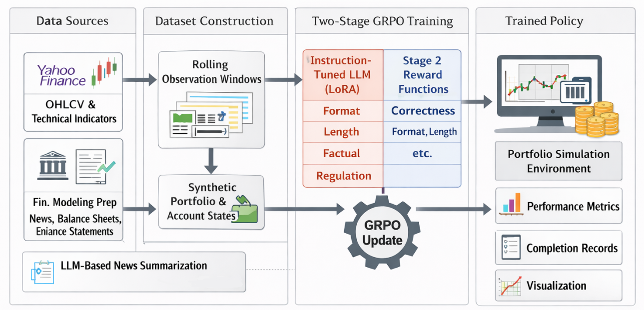

Fig. 1. End-to-end framework. Heterogeneous market data is assembled into rolling observation windows paired with synthetically sampled portfolio states; a fixed system prompt enforces the closed-loop strategy schema; two-stage GRPO fine-tuning optimises a multi-component reward.

fine-tuning with a multi-component reward optimises the policy; and (iv) sequential portfolio simulation evaluates the trained agent on held-out market data. Sampling the portfolio context per-prompt rather than rolling it forward through a single trajectory makes each prompt a self-contained optimisation target that GRPO can score K times in parallel.

B. Closed-Loop Action Space

Each completion is a JSON object with three top-level fields: `reasoning thoughts` (a short rationale that references the observation), `evidence for decision` (the cited signals), and `strategy` (the executable decision). Only the strategy field is consumed by the simulator; the two preceding fields externalise a chain-of-thought trace [20] that is auditable and that the factual-grounding reward can grade.

The strategy object contains six fields: `ticker`, `action` $\in$ {buy, sell, hold}, `quantity` (in shares), `current price`, `target price`, and `stop loss price`. Two design choices distinguish this schema from prior LLM-trading systems. First, every buy must declare its own take-profit and stop-loss before the position is opened, and those values are then attached to the position and govern its exit until the position is closed. The schema requires `target price` and `stop loss price` on hold and sell as well so that the parser remains stateless across actions, but only the buy-side TP/SL is graded by the reward; on existing positions, the regulation reward checks exit-trigger consistency against the portfolio's *stored* TP/SL — the values committed at the original buy — rather than against any newly emitted ones. This makes the closed-loop exit plan a mandatory part of *opening* a position rather than an optional afterthought [6]. Second, `quantity` is denominated as a whole-share count rather than as a percentage of cash or a categorical bucket. A share count is directly executable against the account state, and it lets the reward function compute the notional-to-account ratio $f_{\text{acct}} = q\, p_{\text{curr}}/V_{\text{acct}}$ for tiny-trade detection and well-sized-trade shaping — a signal that a bucketed encoding would flatten away. A thin validation/normalisation layer enforces schema legality and absorbs minor formatting variance; strategies that still fail are scored as bounded failures in training and treated as no-ops at evaluation.

C. Multi-Component Reward

The training signal comprises five components, each bounded to $[-1, +1]$. The *format* reward scores parseable JSON, presence of the reasoning and evidence fields, and schema-legal strategy objects. The *length* reward is a symmetric triangular ramp around a target $L^* = 800$ tokens, peaking at $+1$ when $L = L^*$ and decaying linearly to zero at $L = 0$ and $L = 2L^*$. The *factual grounding* reward sums three terms: hard alignment of the strategy's ticker and current price against the observation, textual reference to named indicators or news, and approximate numeric citation of observed values. The *regulation* reward groups trading-rule penalties into five families — basic sanity, buy-side legality and sizing, sell-side legality and sizing, hold semantics, and take-profit/stop-loss trigger consistency.

The *correctness* reward is the primary profit signal. For every completion, two parallel forward simulations run from the decision day t for $H = 5$ trading days: a do-nothing baseline that walks any held position forward, and an action branch that first executes the strategy and then advances under its updated portfolio. Letting V^{act}_{t+H} and V^{base}_{t+H} denote the two horizon-end account values and V^{acct}_t the decision-day total, the base correctness reward is

$$r^{\mathrm{base}}_t \;=\; \tanh\!\left(\frac{V^{\mathrm{act}}_{t+H} - V^{\mathrm{base}}_{t+H}}{\sigma\, V^{\mathrm{acct}}_t}\right), \tag{1}$$

with $\sigma = 0.02$, which is our expected return for a good trading strategy. That is a scaler to map strategy profit to account value change. Hold is the natural neutral case: identical branches yield $\tanh(0) = 0$, with rank set by GRPO's group comparison.

D. Two-Stage GRPO Fine-Tuning

GRPO [5] is the PPO variant introduced by DeepSeek that replaces the learned value network with an on-the-fly group baseline computed from K parallel rollouts of the same prompt. For each prompt x, we sample $K = 6$ completions $\{y_i\}$ from the current policy, compute group-relative advantages $A_i = (r_i - \mathrm{mean}(r))/(\mathrm{std}(r) + \varepsilon)$ across the weighted reward r_i, and update by maximising the clipped surrogate with a Kullback–Leibler penalty against a frozen reference policy. Full fine-tuning of a 3B backbone is infeasible on the 24 GB development GPU once gradient buffers, optimiser state, and KV cache for K rollouts are accounted for, so we train through Low-Rank Adaptation (LoRA) [21] with $r = 16$, $\alpha = 16$.

The five rewards are not activated simultaneously. Empirically, turning all five on at once leaves correctness gradient-starved: cold-start completions fail the schema gate, the $\tanh$ path of (1) is rarely reached, and group-relative advantages are dominated by structural-error variance. Stage 1 therefore trains under the four shaping rewards at unit weight until they plateau on validation prompts; Stage 2 then promotes correctness to weight 0.60 and demotes the shaping rewards to $\{0.16, 0.08, 0.08, 0.08\}$ for regulation, length, format, and factual grounding. Stage 2 also lowers the learning rate $4\times$ (to 3×10^{-6}) to protect Stage 1 structure and widens sampling (T=0.90, top-p=0.92) so sibling rollouts diverge in quantity; without that diversity the group-relative advantage collapses in the tiny-trade local optimum the correctness reward is designed to penalise.

IV. Experiments

A. Setup

We evaluate on five U.S.-listed equities chosen as one representative per market archetype: AAPL (mega-cap technology), CAT (cyclical), KO (defensive consumer staple), JPM (financial), and GME (high volatility). The dataset covering each ticker's seven-day observation windows is partitioned $80\%/20\%$ chronologically per ticker, and the test split spans the same 101 trading days (2025-11-03 to 2026-03-30) for every method. Stage 1 runs for 500 steps until the four shaping rewards plateau on validation prompts, followed by 1000 Stage 2 steps that cover roughly half an epoch of the training set.

The primary backbone model is Llama-3.2-3B-Instruct; an independent Qwen3-4B-Instruct backbone is included in Sec. IV-B as a cross-architecture check. Each LLM result is averaged over five independent decoding runs ($k = 5$); the DQN baseline is averaged over five training seeds; Buy & Hold is deterministic. We report cumulative return (CR), annualised Sharpe ratio (SR) [22], maximum drawdown (MDD), and *clean rate* — the fraction of model completions that pass schema and rule validation and are therefore eligible for execution.

B. RQ1: Does Pure-RL Fine-Tuning Improve LLM Trading?

We compare the proposed framework against five reference points: each LLM's own pre-RL base, the same base after our two-stage GRPO schedule, the DQN trader of [3] re-run on the test window, and a passive Buy & Hold portfolio that splits the initial capital evenly across the five tickers. Results are summarised in Table I.

GRPO training improves both base models. Llama-3.2-3B gains 0.74 percentage points of CR ($+1.22\% \rightarrow +1.96\%$) and a one-point reduction in MDD; Qwen3-4B gains 0.38 points of CR, more than halves its MDD ($5.66\% \rightarrow 2.59\%$), and lifts its clean rate from 70.6% to 98.7%. Both LLM families dominate the DQN baseline by a wide margin on every metric reported, with DQN losing 7.69% on average and posting the worst Sharpe and the largest drawdown of any agent considered.

Buy & Hold still outperforms every LLM agent on raw CR ($+3.14\%$ versus the best LLM result of $+1.96\%$). The gap is consistent with the structural sizing constraint imposed by our regulation reward, which discourages single-ticker concentration so the LLM portfolio can never take on the full

TABLE I
RQ1 MAIN RESULTS, AVERAGED ACROSS THE FIVE-TICKER TEST WINDOW (2025-11-03 TO 2026-03-30). EACH LLM AND DQN VALUE IS THE MEAN OVER FIVE INDEPENDENT RUNS; BUY & HOLD IS DETERMINISTIC.

Model	CR (%)	SR	MDD (%)	Clean (%)
Llama-3.2-3B base	$+1.22$	-0.28	10.22	30.6
Llama-3.2-3B RL	$\mathbf{+1.96}$	-0.31	9.23	31.4
Qwen3-4B base	$+0.52$	-0.69	5.66	70.6
Qwen3-4B RL	$+0.90$	-0.70	**2.59**	**98.7**
DQN [3]	-7.69	-1.56	13.61	—
Buy & Hold	$+3.14$	$+0.28$	13.09	—

positional exposure that a passive equal-weight buyer takes on day one. The flip side is risk: MDD is uniformly lower for the LLM agents (2.59%–10.22%) than for Buy & Hold (13.09%), so quantity-aware sizing trades absolute return for downside containment. The clean-rate gap between the two LLM families (31.4% vs. 98.7% after RL) is structural rather than training-induced: the dominant Llama failure modes are budget-aware buy quantities and forgetting to hold when fully invested, both of which stem from the 3B base's weak arithmetic competence and which fine-tuning alone cannot teach.

C. RQ2: Position Sizing as a Risk-Control Variable

Sec. IV-B attributed the CR gap between the quantity-aware policy and Buy & Hold to the regulation reward's sizing constraint. We test that attribution directly by removing quantity from the action space and retraining the policy end-to-end. A pure evaluation-time override would mix policy behaviour with a reward–action mismatch (the original model was trained with sizing rewards active), so the ablation must touch every component that quantity touches and leave everything else unchanged.

Concretely, the synthetic portfolio generator is collapsed to a bimodal state: each training observation either holds a full position with available cash near zero, or holds no position with the full account in cash, so the policy always faces an all-in/all-out decision. The system prompt is stripped of the quantity field and its associated trading rules. On the reward side, only the sizing terms (buy and sell tiny-trade ramps, well-sized-trade shaping, conviction multiplier) are removed; format, length, factual grounding, correctness, and the remaining regulation checks (budget legality, ownership, TP/SL trigger consistency, hold semantics) are retained without modification. Hyperparameters, LoRA configuration, the two-stage schedule, and the evaluation window are identical to Sec. IV-B.

Results are summarised in Table II. Removing the quantity dimension closes the profit gap to Buy & Hold and does so conclusively: CR rises from $+1.96\%$ to $+3.17\%$, a gain of 1.21 percentage points that more than fills the 1.18-point deficit reported in Sec. IV-B. The Sharpe ratio crosses zero for the first time across every LLM agent considered in this work, moving from -0.31 to $+0.18$. MDD does increase from 9.23% to 12.19%, consistent with the larger per-trade exposure and approaching Buy & Hold's 13.09%. These three movements together reproduce almost exactly the tradeoff that the regulation-reward sizing terms were designed to enforce: capping trade size trades nominal return for downside containment, and removing the cap recovers the return at the cost of the containment.

TABLE II
RQ2 ACTION-SPACE ABLATION. NO-QUANTITY REPRESENTS FOR REMOVING POSITION SIZING OFF FROM ALL THE PROCESS. NO-QUANTITY IS ABLE TO REACH THE CR OF BUY AND HOLD STRATEGY WITH A SMALLER MDD.

Model	CR (%)	SR	MDD (%)	Clean (%)
Quantity-aware	$+1.96$	-0.31	**9.23**	31.4
No-quantity	$\mathbf{+3.17}$	$\mathbf{+0.18}$	12.19	**73.3**
Buy & Hold	$+3.14$	$+0.28$	13.09	—

The clean-rate jump from 31.4% to 73.3% is consistent with the failure-mode analysis of Sec. IV-B: removing the sizing field eliminates a large class of failures at the schema level rather than through better training. RQ2 therefore characterises explicit per-trade sizing not as a CR concession but as a path-stability lever, with the choice between regimes a portfolio-level risk-return preference rather than a training question.

V. CONCLUSION

We presented a closed-loop GRPO framework that fine-tunes a generic instruction-tuned LLM into a trading agent emitting action, per-trade quantity, take-profit, and stop-loss, on public data with no SFT cold-start. Two empirical findings support the design.

First, pure RL alone converts a generic LLM into a competent, rule-respecting trader. Both Llama-3.2-3B and Qwen3-4B receive measurable CR uplift over their bases and dominate the DQN baseline on every metric. Trading-R1 [4] already showed pure RL improving over a non-RL base; our experiments suggest a closed-loop action space and a multi-component reward can further amplify that contribution, even without any SFT corpus.

Second, per-trade quantity functions as a risk-control variable rather than a performance variable. Removing the sizing field from the action space and retraining end-to-end recovers Buy & Hold parity on cumulative return, but at the cost of an approximately three-percentage-point deeper maximum drawdown. Explicit per-trade sizing therefore trades terminal return for path stability — a risk-management lever folded into the policy itself rather than imposed outside it.

Together, closed-loop action-space design and a multi-component reward stack are sufficient to convert a generic instruction-tuned LLM into an interpretable, rule-respecting trading agent with only reinforcement learning.

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